INCH POUND
MIL-DTL-32193
4 August 2005

DETAIL SPECIFICATION
INITIATOR, EXPLOSIVE ACTUATED, JAU-29/A

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. The initiator, explosive actuated, JAU-29/A converts explosive energy from a shielded mild detonating cord (SMDC) into ballistic gas energy used in the aircrew escape system of the TF-18 and TAV-8B aircraft.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-I-8500	Interchangeability And Replaceability Of Component Parts For Aerospace Vehicles
MIL-P-20444	Primer, Percussion - Parts For And Loading - Assembling And Packing
MIL-D-21625	Design And Evaluation Of Cartridges For Cartridge Actuated Devices

Comments, suggestions, or questions on this document should be addressed to Commander, Indian Head Division, Naval Surface Warfare Center, Technical Information Branch (Code 4230), 101 Strauss Avenue, Indian Head, Maryland 20640-5035, or emailed to amanda.penn@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil/

AMSC N/A FSC 1377

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-464 Electromagnetic Environmental Effects Requirements For Systems

(Copies of these documents are available online at http://assist.daps.dla.mil/ or from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia PA 19111- 5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

DRAWINGS

Naval Air Systems Command

850AS801 Ballistic Test Fixture
850AS803 Vibration Test Fixture

(Application for copies should be addressed to Commanding Officer, Naval Surface Warfare Center, Indian Head, Technical Information Branch (Code 4230), Indian Head, MD 20640.)

2.3 Non-Government publications and drawings. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E1742 Standard Practice for Radiographic Examination

(Copies of this document are available online at http://www.astm.org/ or from the American Society for Testing and Materials Customer Service, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE AS 5604 Compass, Magnetic, Pilots Standby

(Copies of this document are available online at http://www.sae.org/, or from the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, Pennsylvania, United States, 15096-001.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, superseded applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article sample. When more than 12 months have elapsed since production of the latest accepted lot, a first article sample of 60 initiators that have met the requirements of 3.2 through 3.5 shall be required. The sample shall be tested as specified in paragraph 4.3.

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3.2 Design and construction. The JAU-29/A initiator design and construction shall be identical to that subjected to qualification testing and approved for service use.

3.3 Envelope and installation. The dimensions and configuration of the JAU-29/A initiator shall be as shown on DWG 850AS139. All parts having the same vendor's item number shall be functionally and dimensionally interchangeable in accordance with MIL-I-8500.

3.4 Primary components. The M42C1-979 (PA101) percussion primer used in the JAU-29/A initiator shall be manufactured and tested in accordance with MIL-P-20444 (primer, percussion - parts for and loading - assembling and packing). The primers shall be from lots manufactured within a year from the date of cartridge loading. All explosive materials used in the manufacture of this item are considered primary components. Only primary components from a single lot will be used in a cartridge lot.

3.5 Inspection

3.5.1 Cartridge inspection.

3.5.1.1 Visual inspection. All cartridges manufactured under contract shall be visually inspected and screened for defects as outlined in 4.4.

3.5.1.2 Leakage. Cartridges shall not exhibit a leak rate in excess of 1×10^{-5} cubic centimeters per second of air when tested in accordance with 4.5.

3.5.1.3 Radiographic inspection. Cartridges shall be free of imperfections in assembly when examined radiographically in accordance with 4.6.

3.5.2 Initiator inspection.

3.5.2.1 Visual inspection. All initiators manufactured under contract shall be visually inspected and screened for defects as outlined in 4.4.

3.5.2.2 Radiographic inspection. Initiators shall be free of imperfections in assembly when examined radiographically in accordance with 4.6.

3.5.2.3 Demagnetization. Initiators shall not deflect an approved pilot's standby compass more than three degrees in either direction when passed in front of the compass when tested in accordance with 4.7.

3.5.2.4 Electrical bonding resistance. Initiators shall not exhibit an electrical bonding resistance in excess of 0.0025 ohm when tested in accordance with 4.8.

3.6 Ballistic requirements. Initiators from each first Article and production lot shall meet the requirements listed below when subjected to the test prescribed in 4.10.

3.6.1 Maximum pressure. The upper (u) and lower (l) limits for maximum pressure (psig) when fired over the temperature range from –65 °F to +200 °F shall be u = 5100 psig and l = 3850 psig, see 4.10.1.1.

3.6.2 Ignition delay. The limit for ignition delay shall not be greater than 5 milliseconds. See 4.10.1.2.

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3.6.3 Time to maximum pressure. The limit for time to maximum pressure (milliseconds) when fired over the temperature range from –65 °F to +200 °F shall not be greater than 20 milliseconds. See 4.10.1.3.

3.6.4 Misfire. There shall be no misfires. See 4.10.1.4.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First Article inspection (see 4.3).

b. Conformance inspection (see 4.4 through 4.10).

4.2 Sampling. Samples for ballistic test shall be selected from the production lot in accordance with Table I. Reduced sample size is recommended after 10 coinsecutive acceptable lots. Sample plan to be used will be normal unless otherwise specified in the contract. An additional six initiators shall be provided with each test sample for investigative purposes. Sample initiators and initiators made available for investigative purposes shall not be applied as part of the quantity specified for delivery by the contract or order. The production lot size shall be 500 maximum initiators.

Table I. Sample sizes for ballistic test.

Lot Size	Normal Sample Size	Reduced Sample Size
2-8	2	2
9-15	3	2
16-25	5	3
26-50	8	5
51-90	13	5
91-150	20	8
151-280	32	13
281-500	50	20

4.3 First Article sample. Before entering into quantity production, an acceptable first article sample of 60 initiators shall be prepared using the same procedures and methods proposed for the production lot. Fifty-four of these initiators shall be expended in the tests listed herein and six shall be retained for investigative purposes. A first article sample acceptable for functional and environmental testing shall be defined as a sample which has met the requirements of paragraphs 3.2 through 3.5 of this specification. The contractor shall deliver the initiators to the activity designated by the contracting agency for tests. first article sample inspection shall consist of the following tests: temperature and humidity cycling (18), vibration (6) shock (6), –65 °F ± 5 °F (12), and 200 °F ± 5 °F (12). The numbers in parentheses denote the number of initiators allotted for each test. All tests are done as prescribed in MIL-D-21625 except vibration which is prescribed in 4.9 herein. The initiators shall be fired as described in 4.10 and shall meet the functional requirements referenced in 3.6. Any quantity production prior to notification by the contracting agency of first article sample acceptability shall be at the contractor's risk. The first article initiators will not be applied as part of the quantity specified for delivery by the contract.

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4.4 Initiator and cartridge visual inspection. Visually examine all cartridges prior to initiator assembly and assembled initiators. Reject those with inadequate or illegible descriptive markings or serial numbers, dents, deep scratches, or other defects which may prevent proper installation or function.

4.5 Cartridge leak test. Each cartridge produced under contract shall be leak tested in a dry gas leak tester to determine conformance to 3.5.1.2. Units which exhibit a leak rate in excess of 1×10^{-5} std. CC/SEC of air at a pressure differential of 1.0 ± 0.1 atmosphere shall be rejected.

4.6 Radiographic examination.

4.6.1 Cartridge examination. All cartridges shall be arranged on trays or boards in consecutive numerical order and each radiograph shall carry a permanent identification of the cartridge displayed thereon. The cartridges shall be positioned to give the most revealing cross-section of the cartridge. Defective cartridges found by the vendor shall be clearly identified on the radiographic plate and those defective shall be removed from the production lot. Radiographs or copies of the entire production lot shall accompany the ballistic sample to the activity conducting the production lot acceptance test. (see 6.2).

4.6.1.1 X-ray examination. Each cartridge shall be x-rayed, prior to initiator assembly, in accordance with ASTM E1742, to confirm the presence and proper assembly of all parts. Any observable imperfections in assembly shall be cause for cartridge rejection.

4.6.1.2 N-ray examination. Each cartridge shall be n-rayed prior to initiator assembly, to confirm the presence of the propellant non-metal parts, no foreign hydrogenous materials and the proper assembly of pyrotechnic/non-metallic materials. Any observable imperfection in assembly shall be cause for cartridge rejection.

4.6.2 Initiator examination. All initiators shall be identified with serial numbers prior to radiographic examination. The initiators shall be arranged on trays or boards in consecutive numerical order and each radiograph shall carry a permanent identification of the initiators displayed thereon. The initiators shall be positioned to give the most revealing cross-section of the initiator. The radiograph identification shall include the complete lot number, as stamped on the initiators, and the span of serial numbers displayed. Discontinuities in serial numbers shall be identified on the radiographic plate. Defective initiators found by the vendor shall be clearly identified on the radiographic plate and those defective shall be removed from the production lot. Radiographs or copies of the entire production lot shall accompany the ballistic sample to the activity conducting the production lot acceptance test. (see 6.2).

4.6.2.1 X-ray examination. Each assembled initiator shall be x-rayed in accordance with ASTM E1742, to confirm the presence and proper assembly of all parts. Any observable imperfections in assembly shall be cause for initiator rejection.

4.7 Demagnetization. Each initiator shall be moved past a pilot's standby compass face (in accordance with SAE AS5604) by placing the initiator six inches from the compass face, in a north-south horizontal position with the compass set in a east-west heading. The initiator and the compass shall be placed at the same height from the floor. The initiators shall be moved past the compass face in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis, and moved

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back past the compass face. This procedure shall be repeated for each 90 degrees rotation of the initiator. The initiator shall be acceptable if the requirements of 3.5.2.3 are met.

4.8 Electrical bonding resistance. Each initiator shall be tested using a four point probe milliohmmeter in accordance with MIL-STD-464. The electrical bonding resistance shall be measured between the following points on the initiator:

A. The thread of the inlet SMDC port and the thread of the outlet fitting.

B. The mounting lugs and the thread of the outlet fitting.

C. The thread of the inlet SMDC port and the mounting lugs.

The initiator shall be acceptable if the requirements of 3.5.2.4 are met.

4.9 Vibration test.

4.9.1 Non-gunfire vibration. The initiators shall be mounted in the vibration test fixture NAVAIR drawing no. 850AS803 for the JAU-29/A initiator. The non-gunfire vibration test shall be performed on all six initiators mounted on the test fixture at the same time and one initiator shall be monitored for response during the test. The test specimens shall be exposed to the sinusodial vibration environment shown in Figure 1 and Table II with the frequency varying logarithmically from 5 Hz to 2,000 Hz and returning to 5 HZ in approximately 20 minutes. The specimens will be exposed to the environment for one hour in each of the three mutually perpendicular axes, at ambient, +200ºF, and -65ºF, for a total of nine hours of testing.

Table II. Non gunfire vibration test levels.

Frequency (hz)	Acceleration/Amplitude
5 to 20	0.10 inch D.A.
20 to 33	± 2 g's peak
33 to 74	0.036inch D.A.
74 to 2000	±10 g's peak

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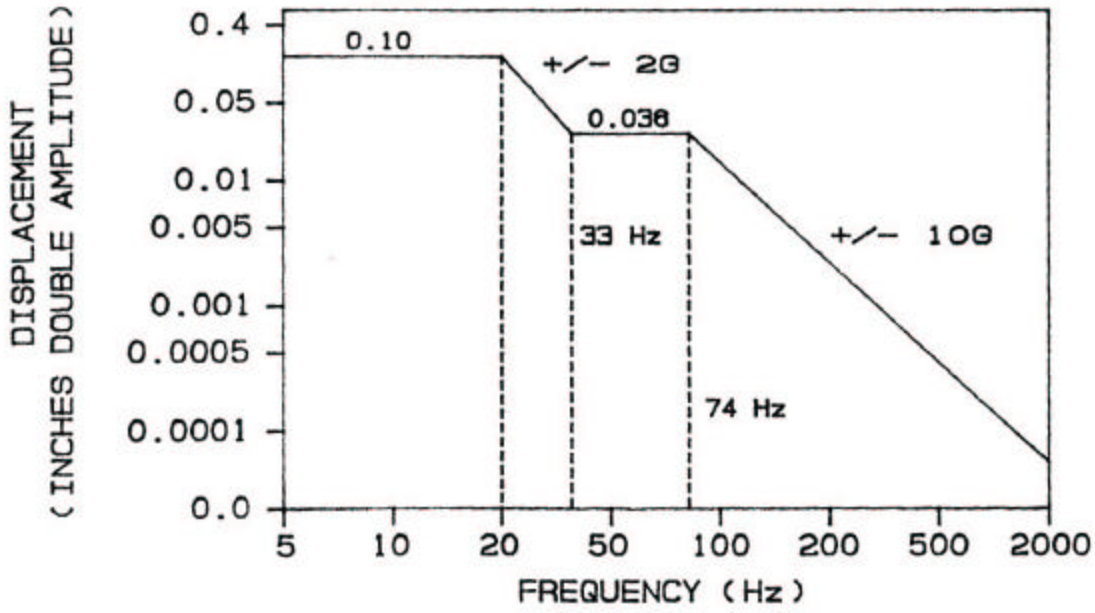


Figure 1. Non gunfire vibration.

4.9.2 Gunfire vibration test.

4.9.2.1 The gunfire vibration test shall be performed on all 6 initiators, on each of the three orthogonal axes, and shall be conducted at + 70 °F test temperature only. The gunfire vibration test shall be performed on all six initiators mounted on the test fixture at the same time and one initiator shall be monitored for response during the test.

4.9.2.2 Resonance search. A resonance survey shall be conducted on each of the six initiators at +70 °F in each of the three orthogonal axes. The frequency sweep shall be made from 50 hz to 500 hz and back to 50 hz in approximately 7.50 minutes to the vibration test levels shown in Table III. The resonance points shall be noted. A resonance point is defined as a frequency which has a transmisibility (output/input) of two or greater. If more than three resonant points are noted from the resonance survey, only the three more severe will be used for resonance dwell vibration.

4.9.2.3 Resonance dwell. The six initiators shall be vibrated in each of the three orthogonal axes at +70 °F for 7.50 minutes at each resonant point (to a maximum of three) obtained during the resonance survey in paragraph 3.9.2.2 herein. The gunfire vibration test levels are shown in Table III. Depending upon the number of resonance frequencies determined in the resonance survey per paragraph 4.9.2.2 herein, the remainder of the sweep time in accordance with Table IV shall be used for vibration cycling per paragraph 4.9.2.4 herein.

4.9.2.4 Vibration cycling. All six initiators shall be subjected to sinusoidal vibration cycling at +70 °F for a period of time as specified in 3.9.2.3 on each of the three orthogonal axes with the frequency varying

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logarithmically from 50 hz to 500 hz and back to 50 hz in approximately 7.50 minutes. The gunfire vibration test levels shall be as shown in Table III.

TABLE III. Gunfire vibration levels.

Frequency (hz)	Acceleration or amplitude
50 – 54	.036 inch da.
54 – 210	± 5 g's
210 – 300	.00222 inch d.a.
300 – 500	+ 10 G's

4.9.2.5 Fixed dwell. After sinusoidal vibration cycling, the initiators shall be subjected to a series of fixed dwells as shown in Table IV. The fixed narrow band dwells shall be performed by sweeping the frequency about ±5% of the specified center frequency. For example, the 100 hz dwell should be performed from 95 hz to 105 hz at the logarithmic sweep rate specified in paragraph 4.9.2.4 herein.

4.9.2.6 Sinusoidal sweep. After completion of the fixed dwell testing, the initiators shall be subjected to seven (7) sinusoidal vibration sweeps in each of the three orthogonal axes. The frequency shall vary logarithmically from 500 hz to 2000 hz and back to 500 hz. The sweep cycling rate shall be as specified in paragraph 4.9.2.4 herein, with an applied input of +/-10 g's.

4.9.2.7 After completion of the vibration sweep specified in paragraph 4.9.2.6 herein, the initiators shall be subjected to radiographic inspection (x-ray) per paragraph 3.5.2.2 herein.

4.9.2.8 After radiographic inspection per paragraph 2.5.2.2 herein, the initiators shall be test fired at ± 70°F in accordance with paragraph 4.10 herein.

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Table IV. Gunfire Sinusoidal Vibration Test Schedule
From 50 Hz to 500 Hz
(Test time in minutes per axis)

Number of Resonances	0	1	2	3
Sweep time	30.00	22.50	15.00	7.50
Resonance dwell time	—	7.50	15.00	22.50
Fixed dwells at the following center frequencies: (fd)				
67 Hz	7.50	7.50	7.50	7.50
100 Hz	7.50	7.50	7.50	7.50
135 Hz	7.50	7.50	7.50	7.50
200 Hz	15.00	15.00	15.00	15.00
267 Hz	7.50	7.50	7.50	7.50
300 Hz	7.50	7.50	7.50	7.50
335 Hz	7.50	7.50	7.50	7.50
400 Hz	15.00	15.00	15.00	15.00
467 Hz	7.50	7.50	7.50	7.50
500 Hz	7.50	7.50	7.50	7.50
Total time each axis	120.00	120.00	120.00	120.00

Note: When an equipment resonance occurs within +5% of a fixed dwell frequency, the fixed dwell period shall be omitted and only the resonance dwell performed. The omitted fixed dwell time shall then be added to the sweep time.

4.10 Ballistic test. Initiators shall be selected from each production lot in accordance with paragraph 4.2. The sample of initiators shall be divided, as nearly as possible, into three equal groups. If the sample of initiators is not equally divisible by three, the excess initiator(s) shall be placed in the group that is to be fired at –65 °F ±5 °F. One group each shall be conditioned for a period of time neither less than 4 hours nor greater than 24 hours at –65 °F ±5 °F and +200 °F ± 5 °F. The other group shall be conditioned for a period of time not less than 4 hours at +70 °F ± 5 °F. The initiators shall be fired in the test fixture NAVAIR drawing no. 850AS801, set up as shown in Figure 2 below, within 3 minutes after removal from temperature conditioning. If any initiator is exposed to the temperature environment for greater than the time allowed or not fired within 3 minutes after removal, it shall be returned to a +70 °F ± 5 °F environment for at least 4 hours and then reexposed to the original environment for not less than 4 hours and not more than 24 hours before test firing. The reexposure may be performed as many times as necessary to complete the required ballistic test. Instrumentation shall be provided to measure and record the initiator maximum pressure (psig); ignition delay (milliseconds); and the time to maximum pressure (milliseconds). The closed bomb shall be thoroughly cleaned after each initiator firing. The production lot shall be accepted if the sample fulfills the criteria in 4.10.1.

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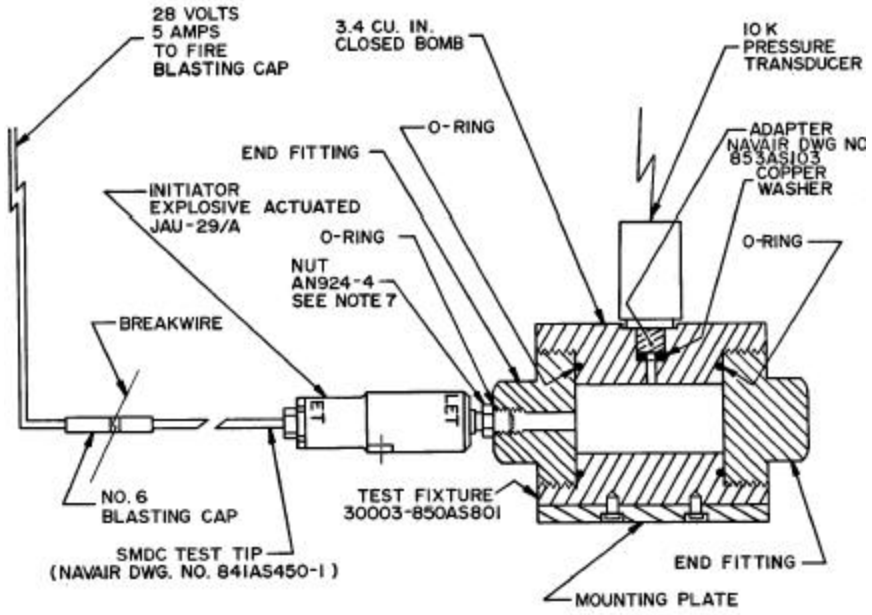


Figure 2. Test Setup.

4.10.1 Acceptance criteria.

4.10.1.1 Maximum pressure. The maximum pressure shall be measured at the point at which the pressure reaches its maximum in the test fixture. The lot shall be acceptable with respect to maximum pressure if the ballistic sample meets the requirements of 3.6.1.

4.10.1.2 Ignition delay. The ignition delay shall be measured from initiation of the inlet SMDC test tip to the first indication of initiator output pressure. The lot shall be acceptable with respect to ignition delay if the ballistic sample meets the requirements of 3.6.2.

4.10.1.3 Time to maximum pressure. The time to maximum pressure small be measured from the first indication of pressure to the point at which the pressure reaches its maximum in the test fixture. The lot shall be acceptable with respect to time to maximum pressure if the ballistic sample meets the requirements of 3.6.3.

4.10.1.4 Misfire. An initiator that does not fire under the conditions specified in 4.10 shall be considered defective and the lot considered unacceptable, unless the misfire is clearly attributable to faulty test equipment or procedure.

4.10.2 Retest. there shall be no retests.

4.10.3 Test failure. If test failure is attributable to an assignable cause, excluding the test initiator, the orginal test results shall be discarded and that test reconducted.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory).

6.1 Intended use. The initiator will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems. The initiator was designed for use in military aircraft and has no commercial application.

6.2 Ordering information. Procurement documents should specify the following:

A. Title, number and date of the source control drawing and this specification
B. Whether first article inspection is required, and if so specify the test activity
C. Production lot size, sample size and test activity
D. Direction for shipping radiographic plates of entire production lot along with ballistic sample (paragraph 4.6) to the activity conducting production lot acceptance tests
E. Items of data required for each first article and production lot

6.3 Safety precautions. The safety precautions requirements of the "contractor's safety manual for ammunition, explosives and related dangerous material" (DOD 4145.26M) are applicable and should be specified in the contract as required by the defence acquisitions regulation (DAR) 1-323. When this specification is used as part of the description of work to be accomplished by a government activity, the safety precaution requirements of "ammunition and explosives ashore" (OP 5) should be made applicable.

6.4 Subject term (key word) listing.

Ejection systems, aircrew
Canopy jettison
SMDC
TF-18

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